ModelIt!

Teacher's Guide

Why Are There Seashells on Mountains?

The Mystery of Fossils Found in Surprising Places

Quick Reference

Grade Level	Grade 4
NGSS Standard	4-ESS1-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: 4-ESS1-1

Identify evidence from patterns in rock formations and fossils in rock layers to support an explanation for changes in a landscape over time.

Three-Dimensional Learning

Science Practice	**Constructing Explanations and Designing Solutions** — Students use fossil evidence and rock layer patterns to construct an explanation for how landscapes change over time.
Disciplinary Core Idea	**ESS1.C The History of Planet Earth** — Local, regional, and global patterns of rock formations reveal changes over time due to Earth forces such as earthquakes and volcanic eruptions.
Crosscutting Concept	**Patterns** — Students identify patterns in rock layers and fossil locations to explain how landscapes have changed over millions of years.

CAST Testing Alignment

MULTIPLE CHOICE:

A hiker finds a fish fossil on top of a mountain. Which explanation best describes how it got there?

CONSTRUCTED RESPONSE:

You find three rock layers on a cliff: the bottom layer has ocean shells, the middle layer has fern leaves, and the top layer has land animal bones. Describe what happened to this area over time.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

The discovery of marine fossils on mountaintops was one of the great mysteries that early geologists had to solve. It is now understood as powerful evidence for the dynamic nature of Earth's surface.

How Fossils Form

Fossils form when an organism dies and is quickly buried by sediment like mud, sand, or volcanic ash. Over thousands to millions of years, the sediment hardens into rock, and minerals replace the original biological material. The most common fossils are from hard parts like shells, bones, and teeth, but soft-bodied organisms can be preserved under exceptional conditions.

Rock Layers Tell a Story

The principle of superposition states that in undisturbed rock layers, the oldest layers are on the bottom and the youngest are on top. Each layer represents a different period in Earth's history. By examining the fossils in each layer, scientists can reconstruct what the environment was like at that time — ocean, swamp, desert, or forest.

How Mountains Rise

Mountains form through tectonic forces — when two plates collide, the crust crumples and folds upward. The Himalayas, for example, contain marine limestone fossils from an ancient ocean floor (the Tethys Sea) that was pushed upward when the Indian plate collided with the Eurasian plate about 50 million years ago. Mount Everest's summit contains marine fossils.

Famous Fossil Locations

The Grand Canyon reveals nearly 2 billion years of rock layers. The Burgess Shale in Canada preserves 500-million-year-old soft-bodied marine creatures. The La Brea Tar Pits in Los Angeles preserve Ice Age mammals. Each location tells a different chapter of Earth's long history through its rock layers and fossils.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify sediment buildup, Earth movement speed, and fossil exposure as the key components of the fossil formation and discovery system.

E **Establish Relationships**
Students determine that sediment buildup positively affects fossil creation (more sediment = more burial = more fossils), and Earth movement speed positively affects fossil exposure (more uplift = more surface exposure).

V **Visualize & Model**
Students build a model showing how sediment layers create fossils underwater, and how Earth movement lifts those layers to where erosion can reveal them.

E **Evaluate Outcomes**
Students run ocean floor and mountain building scenarios to observe how the same fossil can go from deep underwater to a mountaintop.

R **Revise & Extend**
Students add erosion rate or volcanic activity to explore more complex interactions in the fossil exposure system.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"What if I told you the top of a mountain used to be at the bottom of the ocean?"*

Watch For

Show a photo of a seashell fossil found on a mountain. Let students react with wonder.

Slide 2: Learning Objectives

2 min

Say: *"Today we are going to solve one of nature's biggest mysteries."*

Watch For

Have students read objectives aloud. Pre-teach fossil and sediment.

Slide 3: Big Question

3 min

Say: *"Imagine hiking up a mountain and finding a seashell. How did it get there?"*

Watch For

Have students turn and talk with a partner. Collect ideas on the board.

Slide 4: LEVER Framework

2 min

Say: *"We will build a model to figure out this mystery step by step."*

Watch For

Briefly preview each LEVER step. Build excitement for the discovery.

Slide 5: Activity 1: Components

8 min

Say: *"What are the pieces of this puzzle? What has to happen for a fossil to end up on a mountain?"*

Watch For

Guide students through identifying components. Discuss external vs internal.

Slide 6: Activity 2: Connections

7 min

Say: *"More sediment means more things get buried. More Earth movement means more fossils get pushed up. Both are positive!"*

Watch For

Help students draw relationship arrows. Emphasize both relationships are positive.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let us travel millions of years into the past and watch what happens!"*

Watch For

Students predict first, then run scenarios. Compare ocean floor vs mountain building.

Slide 8: Discoveries

4 min

Say: *"Now you know the secret — mountains used to be ocean floors!"*

Watch For

Lead discussion connecting model results to real fossil discoveries like Mount Everest.

Slide 9: STEM Challenge

8 min

Say: *"You are now paleontologists! Plan a dig to find ancient treasures hidden in rock."*

Watch For

Distribute materials. Groups plan their fossil dig using rock layer evidence.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Sediment Buildup, Earth Movement Speed
- **INTERNAL:** Fossil Exposure

Relationships

- **Sediment Buildup to Fossil Exposure:** POSITIVE (+) — More sediment buries more organisms, creating more fossils that can eventually be exposed when layers are lifted up.
- **Earth Movement Speed to Fossil Exposure:** POSITIVE (+) — Faster Earth movement pushes rock layers higher above sea level, where wind and rain can wear away the surface and reveal the fossils trapped inside.

Key Discoveries

- Fossils form when living things get buried in sediment layers on ocean floors or lake bottoms
- Earth movement slowly pushes ocean floor rock layers upward over millions of years, creating mountains
- Once rock layers are lifted high, wind and rain wear away the surface, exposing ancient fossils
- Finding ocean fossils on a mountain is proof that the land was once underwater

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Fossils are just old rocks shaped like animals"	Fossils are the actual preserved remains of once-living organisms, not random rock shapes. The organism died, was buried in sediment, and over millions of years the original material was replaced by minerals, turning it to stone while keeping its exact shape. Scientists can identify species, diet, and even diseases from fossils.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Mountains have always been mountains"	Many mountains were once ocean floors, plains, or even swamps. The Himalayas were a shallow sea 50 million years ago. The Appalachian Mountains were once taller than the Himalayas but have been worn down by hundreds of millions of years of erosion. Earth's surface is constantly changing — just very slowly.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Fossils form quickly"	Fossil formation takes thousands to millions of years. An organism must be buried quickly in sediment before it decomposes, then the sediment must harden into rock over vast time periods. The minerals slowly replace the original material molecule by molecule. This is why fossils are rare — most organisms decompose before they can be preserved.